Variance, Covariance, and Correlation

Terminology review

- Samples and Features
- Tidy Data Matrix

Mean

The sample mean of a feature is

$$\mu_X = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Variance

Definition: The (sample) variance of the feature measurements $x_1, \ldots, x_n$ is

$$\sigma_X^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu_X)^2 = \frac{1}{N}\left(\sum_{i=1}^{N} x_i^2\right) - \mu_X^2$$

$$\sum_{i=1}^{N}(x_i-\mu_x)^2 = \sum_{i=1}^{N} x_i^2 - 2x_i\mu_x + \mu_x^2$$

$$= \sum_1^N x_i^2 - \sum_1^N 2x_i\mu_x + \sum_1^N \mu_x^2$$

$$\sum_1^N x_i^2 - 2\mu_x\sum_1^N x_i + N\mu_x^2$$

$$= \sum x_i^2 - 2N\mu_x^2 + N\mu_x^2$$

$$= \sum x_i^2 - N\mu_x^2$$

Covariance

Definition: If $X = (x_1, \ldots, x_N)$ and $Y = (y_1, \ldots, y_N)$ are two feature vectors then the (sample) covariance is

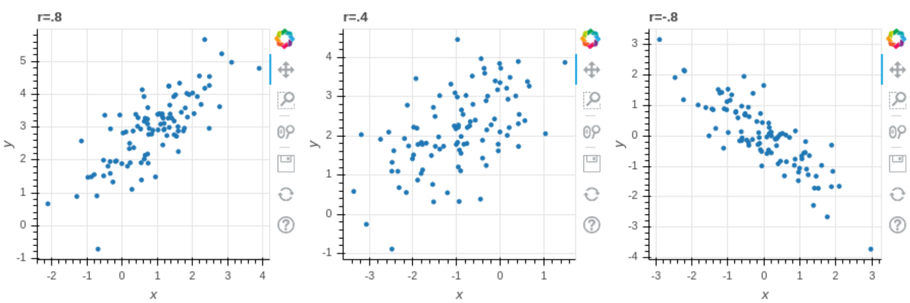
$$\sigma_{XY} = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu_X)(y_i - \mu_Y)$$

Correlation

Definition: Given feature vectors X and Y, the (sample) correlation coefficient r_{XY} is

$$r_{XY} = \frac{\sigma_{XY}}{\sigma_{XX}\sigma_{YY}}$$

$$\sigma_{xx} = \sqrt{\frac{1}{N}\sum x_i^2 - \mu_x^2}$$



The covariance matrix

Definition: Let X be an $N \times k$ data matrix, and let X_0 be its centered version. The (sample) covariance matrix is the $k \times k$ symmetric matrix

$$D_0 = \frac{1}{N} X_0^{\intercal} X_0.$$

Lin Regression

$X^T(Y - X\mu) = 0$

$X^T Y = \underbrace{X^T X}\mu$

$\mu = (X^T X)^{-1} X^T Y$

$X_0 \longleftrightarrow$ N rows $\longleftrightarrow$ samples

k columns $\longleftrightarrow$ features

arranged that all columns sum to zero

$X \rightsquigarrow$ subtract average of each column from every entry in that column

X_0^T $k \times N$

X_0 $N \times K$ $\rightsquigarrow X_0^T X_0$ $K \times K$

$$X_0^T X = \begin{pmatrix} \sum_{i=1}^{N} x_{i1}^2 = \sigma_{x_1}^2 & \sigma_{x_1 x_2} & \cdots \\ & & \\ & & \end{pmatrix}$$

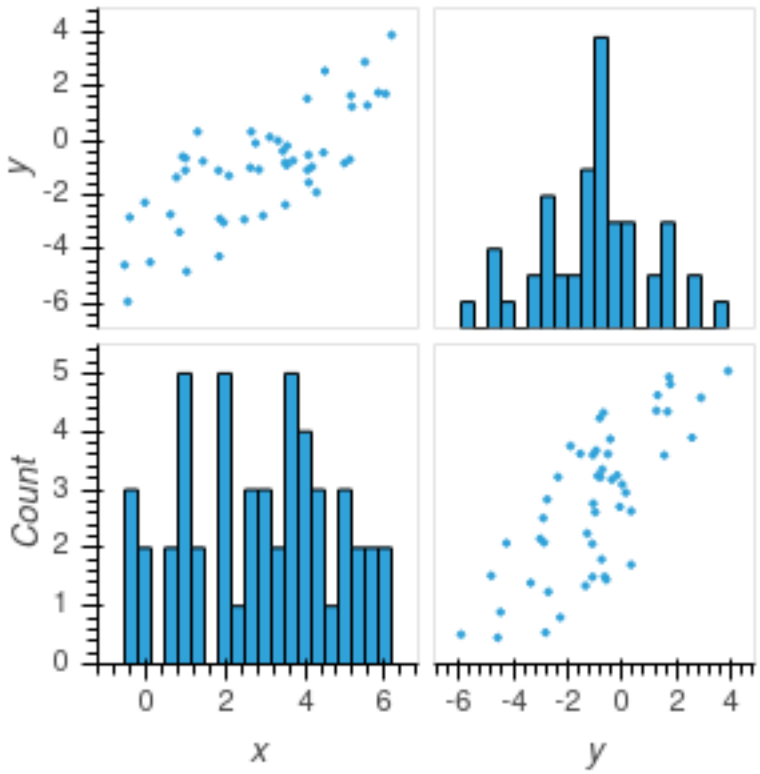
$$X_0 \quad \begin{pmatrix} x_{11} & x_{12} & x_{13} & \cdot & x_{1k} \\ x_{21} & x_{22} & x_{23} \cdots & & x_{2k} \\ & & \vdots & & \\ x_{N1} & \cdot & \cdot & - \ \cdot & x_{Nk} \end{pmatrix}$$

$$X_0^T = \begin{pmatrix} x_{11} & x_{21} & \cdots & x_{N1} \\ x_{12} & x_{22} & & x_{N2} \\ \vdots & & & \\ x_{1k} & x_{2k} & & x_{Nk} \end{pmatrix}$$

i,j entry of $\frac{1}{N} X_0^T X$ = is covariance of feature i with feature j (or variance of $i = j$)

i,j entry = j,i entry

Visualizing the covariance matrix



$\sigma_x^2 \quad \sigma_{xy}$

$\sigma_{yx} \quad \sigma_y^2$